



EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2022
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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CHEMISTRY

Paper 2 Structured Questions

9729/02

15 September 2022

2 hours

Candidates answer on the Question Paper

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, civics group, index number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue, or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Paper 2	
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Total	/ 75

This document consists of **19** printed pages and **1** blank page.

- 1 The carbon family consists of the elements of Group 14. The elements at the top of the group, carbon to germanium, have very different properties from those at the bottom, tin and lead. For instance, Group 14 elements tend to adopt oxidation states of +4, whereas the heavier elements, such as tin and lead, exhibit the +2 oxidation state due to the inert pair effect.

- (a) State the valence shell configuration of Group 14 elements.

$ns^2 np^2$ [1]

- (b) One of the contributing factors to the inert pair effect is the unexpected increase in the ionisation energies, after lead, down the group.

Explain why the 1st ionisation energies are expected to decrease down the group.

Down the group, nuclear charge increases and shielding effect increases. Valence electrons are further away from the nucleus and this leads to decrease in electrostatic forces of attraction between the nucleus and the valence electron. Less energy needed to remove the valence electron, resulting in decrease in 1st I.E. down the group. [2]

- (c) Carbon forms carbide anion, C_2^{2-} in calcium carbide while silicon mostly forms Si^{4+} ions.

- (i) Draw the dot-and-cross diagram of the carbide anion, C_2^{2-} .



[1]

- (ii) In a particular experimental set-up, a beam of $^{28}Si^{4+}$ ions was deflected by an angle of $+4.2^\circ$. Assuming an identical set of conditions, by what angle will the $^{12}C_2^{2-}$ ions be deflected?

$^{28}Si^{4+}$ whose $\frac{q}{m} = \frac{+4}{28} = +\frac{1}{7}$ gives an angle of deflection, θ , of $+4.2^\circ$

$$\theta = k \left(\frac{q}{m} \right) \Rightarrow +4.2^\circ = k \left(+\frac{1}{7} \right) \Rightarrow k = 29.4$$

For $^{12}C_2^{2-}$ whose $\frac{q}{m} = \frac{-2}{24} = -\frac{1}{12}$,

$$\text{angle of deflection, } \theta = 29.4 \times \left(-\frac{1}{12} \right) = \underline{-2.45^\circ}$$

[1]

- (d) Table 1.1 shows that the melting points of the elements of Group 14.

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Table 1.1

element	C	Si	Ge	Sn	Pb
melting point / °C	>3550	1410	937	232	327

Carbon, silicon and germanium each form a solid with the same type of structure. Using bonding and structure, suggest why the melting points of these elements decrease from carbon to germanium.

Carbon, silicon and germanium are of giant covalent structure. To melt these elements, strong covalent bonds between the atoms have to be broken. From C to Ge, the orbitals become bigger and more diffuse. The overlap of the orbitals becomes less effective. Bond strength decreases and hence less energy is required for melting. [2]

- (e) Carbon forms many allotropes such as graphite and diamond. Recent scientific research has found that replacing the graphite electrodes with graphene in lithium-ion batteries can extend battery life.

- (i) Graphene is a **single, one atom thick** layer of graphite. Describe the hybridisation of the orbitals in, and the bonds between, the carbon atoms within graphene.

Each carbon atom is sp² hybridised, and forms 3 σ bonds via head-on overlap between sp² hybrid orbital from an adjacent C atom. The π bond is formed through sideways overlap of the unhybridised p orbitals from an adjacent C atom. [2]

- (ii) Using your answer from (e)(i), explain why graphene can conduct electricity along the plane.

The remaining 2p electron on each carbon atom can delocalise along the plane when a potential difference is applied. [1]

- (iii) State and explain how you would expect the bond strength of carbon-carbon bonds in graphene to differ when compared with that in diamond.

The carbon-carbon bond in graphene is **stronger** than that in diamond. This is due to the **presence of both σ and π bonds / partial double bond character in the C–C bond in graphene, while there are only σ bonds present in diamond.**

[2]

- (f) Diamond and graphite are allotropes of carbon. Fig. 1.1 gives the structure of diamond and graphite. Diamond is an important abrasive while graphite is used as a lubricant.

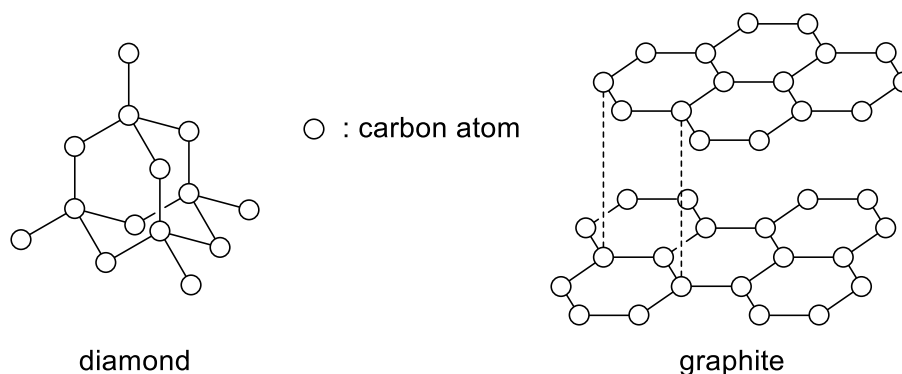


Fig. 1.1

- (i) Explain why the sign of the entropy change for the conversion of diamond to graphite is positive.

There is greater disorderliness as the **layers of graphite are loosely held by weak instantaneous dipole-induced dipole interactions** as compared to the carbon atoms being held in the 3-dimensional lattice in diamond. This results in **greater spread of energies and more ways to arrange the particles in graphite compared to diamond.**

[2]

- (ii) The standard enthalpy change of reaction for the conversion of diamond to graphite is exothermic. Explain whether the conversion of diamond to graphite is spontaneous at all temperatures.

Since $\Delta S > 0$, $-T\Delta S$ is always negative at all temperatures.

and $\Delta H < 0$ (given in question). Hence $\Delta G = \Delta H - T\Delta S \Rightarrow \Delta G < 0$ is spontaneous at all temperatures.

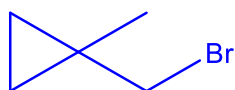
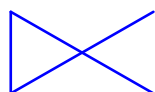
[2]

- (iii) In daily life, we do not observe diamond converting to graphite readily. Explain why this is so.

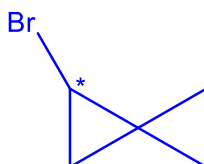
Although the reaction is spontaneous, it has a high activation energy and it takes place at a very slow rate.

[1]

- (g) A three-membered ring cycloalkane, C_5H_{10} , gives only two mono-brominated products with Br_2 under uv light. Only one of the two mono-brominated product is chiral. Give the structural formulae of the cycloalkane and the two mono-brominated products formed. State the ratio in which the two mono-brominated products will be formed.



bromoalkane A



bromoalkane B

ratio of the products A : B = 6 : 4 = 3 : 2

[3]

[Total: 20]

- 2 (a) The reaction of secondary halogenoalkanes with NaOH(aq) may be a first order or second order reaction.

0.0500 mol dm⁻³ (C₆H₅)CHBrCH₃ was reacted with 2.00 mol dm⁻³ NaOH(aq) at a constant temperature and the half-life is constant at 277 seconds.

- (i) Explain the meaning of the following terms:

- order of reaction
- half-life

The order of reaction with respect to a given reactant is the power to which the concentration of that reactant is raised to in the rate equation.

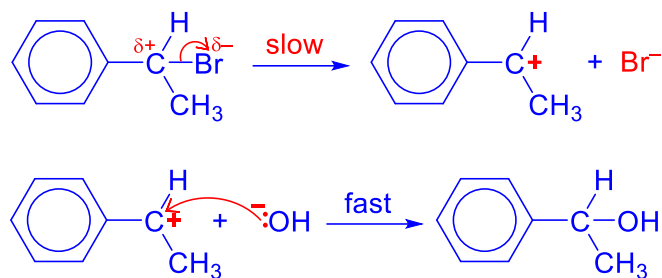
The half-life of a reaction is the time taken for the concentration of a reactant to decrease to half its initial value.

..... [2]

- (ii) Although the half-life of the experimental study is constant, explain why the half-life of the reaction cannot be used to determine whether the reaction takes place via an S_N1 or S_N2 mechanism.

As [NaOH(aq)] is significantly higher than/ in large excess of [(C₆H₅)CHBrCH₃], the [NaOH(aq)] effectively remains constant such that the reaction would be approximated as first-order kinetics/ would behave as a pseudo 1st order reaction even if was 2nd order kinetics. Hence, it is not possible to differentiate between S_N1 (first-order kinetics) and S_N2 (second-order kinetics) based on the half-life of the experimental study. [1]

- (iii) Optically pure $(\text{C}_6\text{H}_5)\text{CHBrCH}_3$ reacts with NaOH via an $\text{S}_{\text{N}}1$ mechanism. Describe the mechanism. In your answer, show any relevant charges, dipoles or lone pairs of electrons you consider important in the mechanism.



[2]

- (iv) Suggest, with reference to the mechanism in (a)(iii), why the product does not rotate plane polarised light.

The carbocation intermediate formed in the rate-determining step has a trigonal planar shape around the carbon centre. The OH^- nucleophile can attack from above and below the plane with equal probability to form an equal amount of enantiomers, thus forming a racemic mixture.

[2]

- (v) With reference to the mechanism in (a)(iii), write the rate equation for the reaction between $(\text{C}_6\text{H}_5)\text{CHBrCH}_3$ and $\text{NaOH}(\text{aq})$. Hence, calculate the rate constant.

$$\text{rate} = k[(\text{C}_6\text{H}_5)\text{CHBrCH}_3]$$

$$t_{1/2} = \frac{\ln 2}{k}$$

$$k = \frac{\ln 2}{t_{1/2}}$$

$$= \frac{\ln 2}{277 \text{ s}}$$

$$= \underline{2.50 \times 10^{-3} \text{ s}^{-1}} \text{ (to 3 s.f.)}$$

[2]

(vi) Suggest and explain how the rate of reaction would change if

- the temperature of the reaction was increased

The reaction will proceed at a faster rate as a greater fraction (or number) of particles will have energy greater or equal to the activation energy of the reaction. Furthermore, the particles move faster, which results in an increase in frequency of collisions, and hence resulting in an increase in the frequency of effective collisions.

[2]

- $(\text{C}_6\text{H}_5)\text{CHBrCH}_3$ was changed to $(\text{C}_6\text{H}_5)\text{CHClCH}_3$ of the same concentration.

Since Br is larger than Cl, the orbital overlap between the halogen and carbon atoms is less effective, causing the bond energy of C–Br to be lower than that of C–Cl and requiring less energy to cleave. Thus, the activation energy increases for the reaction involving $(\text{C}_6\text{H}_5)\text{CHClCH}_3$, resulting in a lower rate constant and hence lowered rate of reaction for the reaction.

[2]

(vii) Water is a more polar solvent than ethanol. With reference to the mechanism in (a)(iii), explain why the rate of $\text{S}_{\text{N}}1$ substitution is lowered when $\text{NaOH}(\text{aq})$ is replaced by ethanolic NaOH .

As ethanol is less polar than water, it is less able to stabilise the carbocation intermediate formed in the rate-determining step, increasing the activation energy of the slow step, which results in a slower rate of formation.

[1]

- (b) The Haber process uses atmospheric nitrogen and hydrogen to produce ammonia gas. When a mixture comprising 1.0 mol of nitrogen and 3.0 mol of hydrogen initially was left to equilibrate in a closed vessel maintained at 500 °C and 250 atm, 0.70 mol of ammonia was attained when the system reached equilibrium.

- (i) Calculate the K_p for the following equilibrium.

	$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$		
	N₂	+	3H₂ ⇌ 2NH₃
initial amt / mol	1.0		3.0 0
change in amt / mol	-0.35		-0.35×3 +0.70
equilibrium amt / mol	0.65		1.95 0.70
total amount of gas at equilibrium: 0.65 + 1.95 + 0.70 = 3.30 mol			

$$K_p = \frac{(p_{\text{NH}_3})^2}{p_{\text{N}_2} \times (p_{\text{H}_2})^3} = \frac{\left(\frac{0.70}{3.30} \times 250\right)^2}{\left(\frac{0.65}{3.30} \times 250\right) \times \left(\frac{1.95}{3.30} \times 250\right)^3}$$

$$= 1.771 \times 10^{-5} \text{ atm}^{-2}$$

$$= \underline{1.77 \times 10^{-5} \text{ atm}^{-2}}$$

[3]

- (ii) Hence or otherwise, calculate the K_p for the following equilibrium.



$$K_p = \frac{(p_{\text{N}_2})^2 \times (p_{\text{H}_2})^6}{(p_{\text{NH}_3})^4} = \left(\frac{1}{1.771 \times 10^{-5} \text{ atm}^{-2}}\right)^2$$

$$= \underline{3.19 \times 10^9 \text{ atm}^4}$$

[1]

- (iii) Helium gas was injected into the equilibrium mixture under constant volume. Explain how the amount of ammonia would change.

The amount of ammonia would not change, as there is no change to the the partial pressures of the reactant gases and products.

[2]

[Total: 20]

- 3 For redox reactions in acidic medium under standard conditions, their $E^\ominus$ values from the *Data Booklet* are valid only at pH 0. However, the pH in typical biological systems is usually around 7. Hence, a new type of reduction potential specific to pH 7 has been defined, which is known as formal reduction potential, $E^\ominus_{\text{formal}}$.

The $E^\ominus$ and $E^\ominus_{\text{formal}}$ values for some important reactions are given in Table 3.1. Reactions which $E^\ominus$ values have not been experimentally determined are not indicated.

Table 3.1

	half-equation	$E^\ominus / \text{V}$	$E^\ominus_{\text{formal}} / \text{V}$
1	$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	+1.23	+0.82
2	oxaloacetate + $2\text{H}^+ + 2\text{e}^- \rightleftharpoons$ malate	+0.33	-0.17
3	$\text{NAD}^+ + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{NADH} + \text{H}^+$	-0.11	-0.32
4	$\text{FAD} + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{FADH}_2$	—	-0.22

- (a) Based on the data in Table 3.1, explain why the $E^\ominus_{\text{formal}}$ is less positive than $E^\ominus$ for the reduction of O_2 gas.

$E^\ominus_{\text{formal}}$ is measured at pH higher than 0. Hence, $[\text{H}^+]$ is lower than 1 mol dm^{-3} ,

position of equilibrium shifts to the left, resulting in a less positive value.

..... [1]

Fig. 3.1 shows the schematic diagram of the citric acid cycle that involves some application of reactions given in Table 3.1. All reactants and products involved in the citric acid cycle are in the aqueous state.

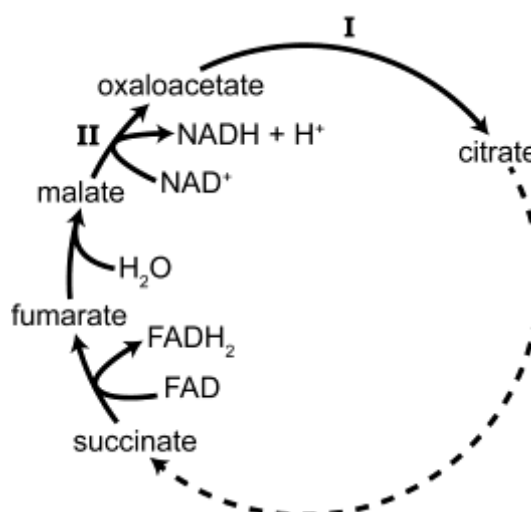


Fig. 3.1

- (b) FAD and NAD^+ are two oxidising agents used in the citric acid cycle. With reference to the $E^\ominus_{\text{formal}}$ values in Table 3.1, explain whether FAD or NAD^+ is a better oxidising agent.

The $E_{\text{formal}}^{\oplus}$ for the reduction of FAD is less negative (or more positive) than the reduction of NAD^+ , hence FAD is a better oxidising agent.

[1]

(c) Calculate $\Delta G_{\text{formal}}^{\oplus}$ for step II.

$$E_{\text{cell, formal}}^{\oplus} = E_{\text{red, formal}}^{\oplus} - E_{\text{oxd, formal}}^{\oplus} = -0.32 - (-0.17) = -0.15 \text{ V}$$

$$\begin{aligned} \Delta G_{\text{formal}}^{\oplus} &= -2 \times 96500 \times (-0.15) \\ &= \underline{+29.0 \text{ kJ mol}^{-1}} \end{aligned}$$

[2]

(d) In the citric acid cycle, it is noted that the consumption of oxaloacetate in step I occurs at a much faster rate than the production of oxaloacetate in step II.

(i) State the effect on the concentration of oxaloacetate due to the difference in reaction rates of the 2 steps.

The concentration of oxaloacetate remains low OR close to zero. [1]

[1]

(ii) Based on your answers in (c) and (d)(i), suggest how this can cause step II to become more spontaneous.

As the concentration of oxaloacetate remains low making

$E_{\text{formal}}^{\oplus}$ (oxaloacetate|malate) more negative, the $E_{\text{cell, formal}}^{\oplus}$ of step II

becomes more positive, thus making the reaction more spontaneous.

[1]

- (e) A student was tasked to electroplate the surface of a steel tank with copper metal using the set-up in Fig. 3.2. A galvanic cell to generate the electricity is required for the electroplating process.

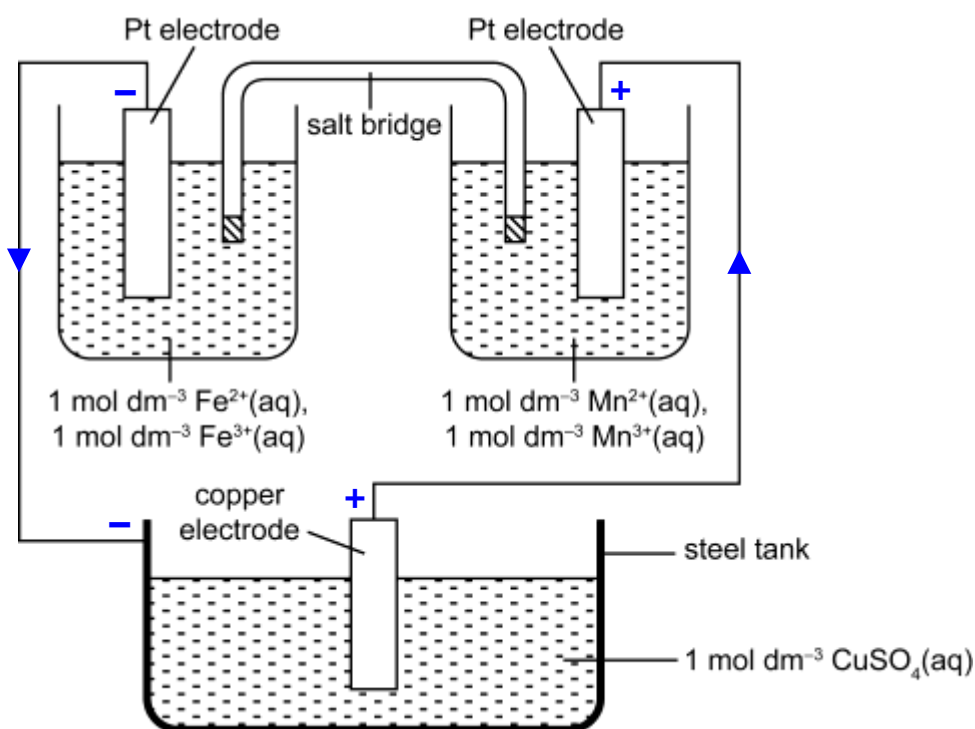


Fig. 3.2

- (i) With reference to the *Data Booklet*, construct an equation for the overall reaction in the galvanic cell.



- (ii) Indicate the polarity of the electrodes in both the galvanic and electrolytic cells and the direction of electron flow between the galvanic and electrolytic cells in Fig. 3.2 [3]

- (iii) Explain why the electrolytes used in the galvanic cell need to be separated into two half-cells in Fig. 3.2.

If the contents of the two half-cells in the galvanic cell were mixed, then a spontaneous reaction would occur without any electrons flowing into the electrolytic cell.

..... [1]

- (f) Celestine is a blue mineral that contains SrSO_4 . When SrSO_4 is heated strongly, it decomposes to form a metal oxide and releases two different colourless gases. One gas rekindles a glowing splint, while the other is SO_2 .

- (i) Construct the balanced equation, including state symbols, for the thermal decomposition of SrSO_4 .



- (ii) Explain why SrSO_4 decomposes at a higher temperature than MgSO_4 .

Sr^{2+} and Mg^{2+} both have same charge. Since Sr^{2+} has bigger ionic size, its charge density is lower. Thus, with the lower polarising power, Sr^{2+} weakens the intramolecular covalent bonds of the anion to a lower extent, leading to stronger bonds to be overcome. Hence, SrSO_4 decomposes at a higher temperature. [2]

- (iii) 5.00 g of SrSO_4 and MgSO_4 were separately heated strongly using the same heat source for an extended period of time. Fig. 3.3 shows how the volume of gas varies with time of heating for MgSO_4 . Sketch on Fig. 3.3 the graph of total volume of gas evolved against time of heating for SrSO_4 .

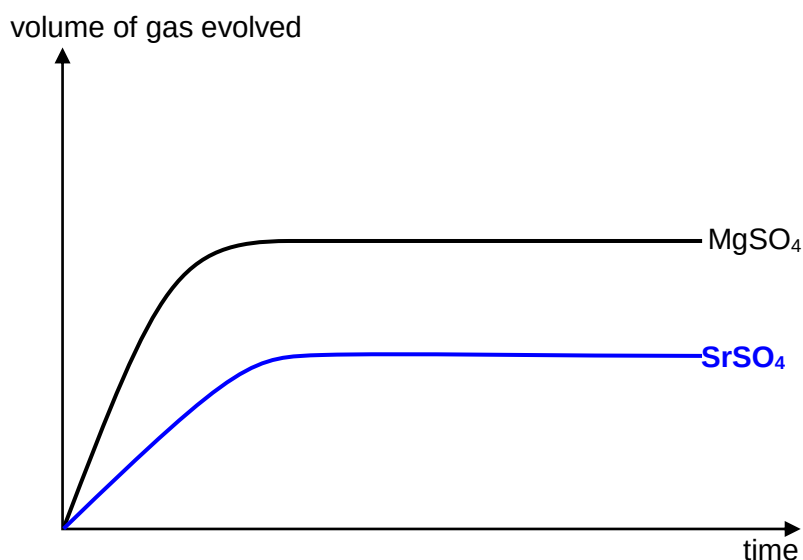


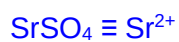
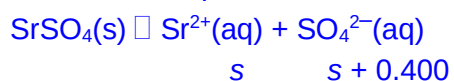
Fig. 3.3

[2]

- (iv) When 2.15×10^{-8} mol of SrSO_4 powder is added to 25.0 cm^3 of $0.400 \text{ mol dm}^{-3}$ $\text{K}_2\text{SO}_4(\text{aq})$ at 298 K, the solution of SrSO_4 just reaches saturation with all the SrSO_4 dissolved. Calculate the K_{sp} of SrSO_4 and state its units.

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Based on the question, all the $\text{SrSO}_4(\text{s})$ has dissolved.



$$[\text{Sr}^{2+}] = (2.15 \times 10^{-8})(1000/25.0) = s$$

$$= 8.60 \times 10^{-7} \text{ mol dm}^{-3}$$

$$K_{\text{sp}} = s(s + 0.400)$$

$$= (8.60 \times 10^{-7})(8.60 \times 10^{-7} + 0.400)$$

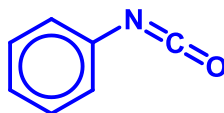
$$= \underline{\underline{3.44 \times 10^{-7} \text{ mol}^2 \text{ dm}^{-6}}}$$

[2]

[Total: 18]

- 4 Phenylamine, $\text{C}_6\text{H}_5\text{NH}_2$, is the simplest aromatic amine. Its main use is in the manufacture of phenyl isocyanate, $\text{C}_6\text{H}_5\text{NCO}$, a precursor to polyurethane, dyes, and other industrial chemicals.

- (a) The carbon atom of the isocyanate group has a similar shape as that in CO_2 . Draw the displayed formula of phenyl isocyanate, $\text{C}_6\text{H}_5\text{NCO}$.



[1]

- (b) The shape of the nitrogen atom of phenyl isocyanate is bent. Hence, state the hybridisation state of the nitrogen atom of phenyl isocyanate. Suggest a bond angle for this nitrogen atom.

There are 3 regions of electron densities around the N atom.

Hence, the hybridisation state is sp^2 .

Bond angle is 118°

[2]

- (c) At 380°C , 1.00 g of gaseous phenylamine has a measured pressure of 950 kPa and volume of $5.90 \times 10^{-5}\text{ m}^3$.

- (i) Assume ideal behaviour, show the molar mass of phenylamine is 96.8 g mol^{-1} .

assume ideal behaviour,

$$pV = nRT = \frac{m}{M}RT$$

$$M = \frac{mRT}{pV} = \frac{1.00 \times 8.31 \times (380 + 273)}{(950 \times 10^3) \times (5.90 \times 10^{-5})}$$

$$= 96.8\text{ g mol}^{-1}$$

[1]

- (ii) The actual relative molecular mass of phenylamine is 93.0 . As all the data are measured accurately, explain why the experimental calculated molar mass in (c)(i) is higher than the actual value.

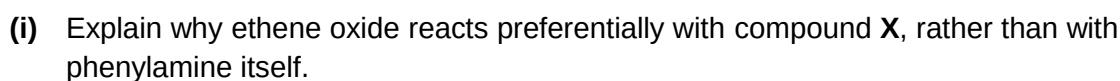
Gaseous phenylamine is actually non-ideal because there is hydrogen bonding

between the phenylamine molecules causing the pV to be smaller than the ideal

pV values, leading to a larger calculated molar mass.

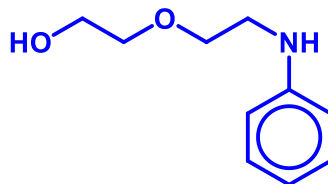
[1]

- Compound **X** that is first formed reacts further with unreacted ethene oxide, giving compound **Y** as the major product.



..... [2]

- Compounds **Y** and **Z** are constitutional isomers. Suggest the structure of compound **Z** and explain why compound **Z** is not formed in significant quantities as compared to compound **Y**.

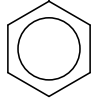
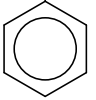
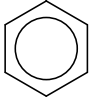


Hence, the O atom in compound X is less likely to donate its lone pair of electrons compared to the N atom. Thus compound Z is not formed in significant quantities. [2]

- (e) The reactivity of three nitrogen-containing compounds with aqueous bromine is shown in Table 4.1

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Table 4.1

	CH_2NH_2  benzylamine	NHCOCH_3  acetanilide	NH_2  phenylamine
reactivity with $\text{Br}_2(\text{aq})$ at room temperature	no reaction	mono-substitution of benzene ring	tri-substitution of benzene ring

- (i) Explain the difference in the reactivity of the three compounds with $\text{Br}_2(\text{aq})$.

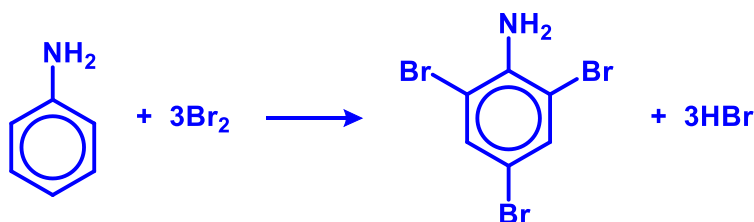
Phenylamine is the most reactive as the lone pair of electrons on N can delocalise into the benzene ring due to overlap between p orbital of N and π electron cloud of benzene. This increases the electron density of the benzene ring greatly, making it most susceptible to electrophilic attack.

Acetanilide is less reactive than phenylamine because delocalisation of the lone pair of electrons on N into the C=O reduces the extent of delocalisation of the lone pair into the benzene ring. Hence, the benzene ring is less electron-rich than phenylamine.

Benzylamine is the least reactive as the lone pair of electrons on N cannot delocalise into the benzene ring, causing the benzene ring to be the least electron-rich.

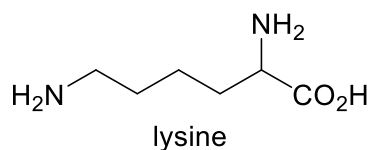
..... [3]

- (ii) Write a balanced equation for the reaction between phenylamine and $\text{Br}_2(\text{aq})$ at room temperature.



[1]

- (f) Lysine is an α -amino acid that is used in the biosynthesis of proteins.



It is an essential amino acid because human body cannot make it, so it must be obtained from food.

The pK_b values of lysine are 11.8, 5.1 and 3.5, with the α -amino group being the weaker base between the two amino groups.

- (i) The fully deprotonated form of lysine is shown in Fig. 4.1. Fill in the pK_b values in the respective boxes.

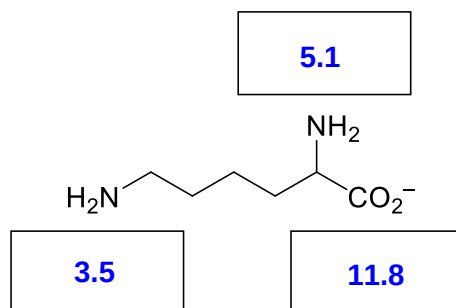


Fig. 4.1

[1]

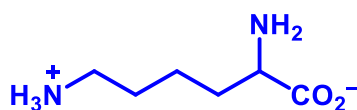
- (ii) Suggest a reason why the α -amino group of lysine is less basic than the amino group on the side chain.

Due to the electron-withdrawing $-\text{COOH}$ group, the lone pair of electrons on the nitrogen of the α -amino group is less available to act as a base.

.....

..... [1]

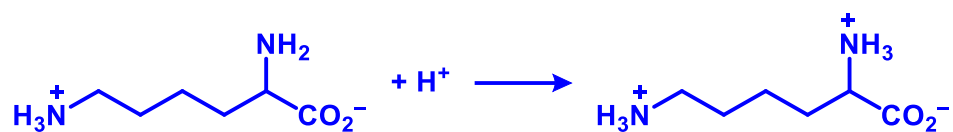
- (iii) Draw the zwitterionic form of lysine and state the pH range at which the zwitterion will exist as the major species.



pH range: 8.9 – 10.5 [1]

- (iv) The zwitterion of lysine can act as a buffer agent in the blood. Write an equation showing how the zwitterion regulate the pH when a small amount of acid is added.

For
Examiner's
Use



[1]

[Total: 17]

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